can see that CoT-decoding is the only decoding strategy that effectively enables language models to reason, while some of the decoding methods even hurt model reasoning compared to greedy decoding.

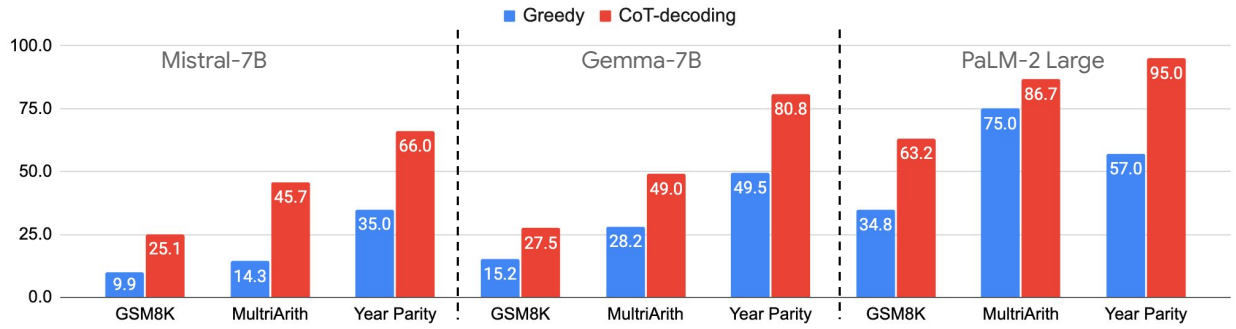

Figure 3 | CoT-decoding effectively elicits reasoning across multiple language model families including PaLM-2, Mistral and Gemma, with significant accuracy gains over three reasoning tasks.

CoT-decoding effectively elicits reasoning across language models. In Figure 3, we show that across three language model families, PaLM-2, Mistral and Gemma, CoT-decoding effectively elicits model's reasoning, yielding consistent accuracy gains over both math and commonsense reasoning tasks, sometimes doubling or even tripling the performance compared to greedy decoding.

CoT-decoding elicits reasoning across model scales. In Figure 4, we show that CoT-decoding enhances reasoning across different model scales over the PaLM-2 model family. On GSM8K, CoT-decoding consistently yields +10-30% absolute accuracy gains. On year parity, when using greedy decoding, the model's performance remains flat even after scaling up model sizes, consistent with the findings in (Allen-Zhu and Li, 2023). In contrast, CoT-decoding significantly boosts the performance by recovering the CoT paths, achieving almost perfect accuracy at larger model scales.

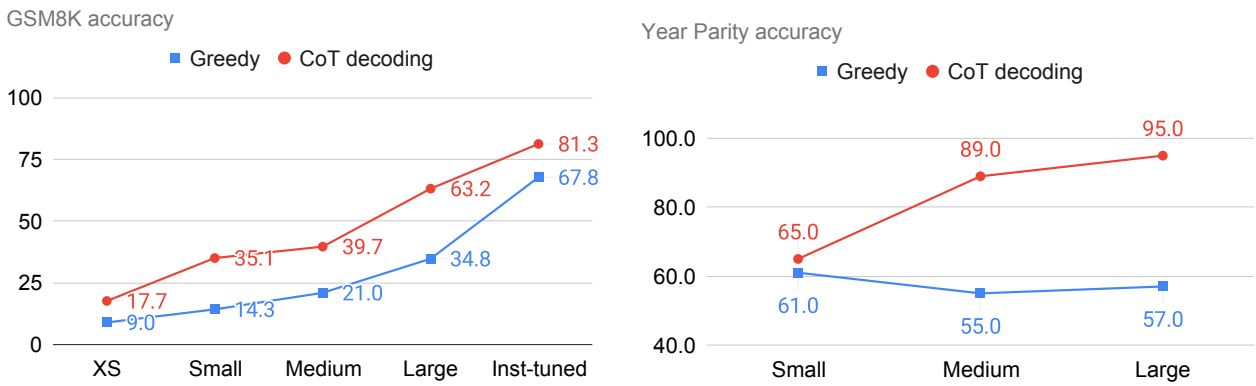

Figure 4 | CoT-decoding reliably improves reasoning performance across model scales (PaLM-2), even when the task does not naturally improve by scaling up only (e.g., year parity).

CoT-decoding partially closes the reasoning gap between pre-trained and instruction-tuned models, without using any supervised data. Intriguingly, we observe that CoT-decoding enables a pre-trained model to achieve a similar performance of an instruction-tuned model: in Figure 4 (left), CoT-decoding achieves 63.2% accuracy on the pre-trained PaLM-2 Large model, close to the performance of the instruction-tuned model of the same scale at 67.8%. The results demonstrate that instruction-tuning with sufficient CoT data (Chung et al., 2022) can be partially achieved by modifying the decoding procedure within pre-trained models.

More interestingly, we observe that CoT-decoding can further improve the instruction-tuned model (Figure 4 (left) and Table 5). The instruction-tuning procedure (Chung et al., 2022) has already incorporated abundant CoT annotations during the fine-tuning process. Consequently, the model is expected to inherently generate CoT paths when addressing reasoning tasks. However, upon analyzing specific examples, we found that even after instruction-tuning, the model occasionally persists in attempting to directly address a question. In contrast, CoT-decoding can enhance the exploration of alternative paths by triggering a CoT first, consequently leading to more accurate answers.

Table 5 | CoT-decoding improves both pre-trained and instruction-tuned Mistral-7B models.

		Pre-trained	Inst-tuned
GSM8K	Greedy	9.9	31.2
	CoT-decoding	**25.1** (+15.2)	**38.2** (+7.0)
MultiArith	Greedy	14.3	37.8
	CoT-decoding	**45.7** (+31.4)	**66.5** (+28.7)
Year Parity	Greedy	35.0	62.2
	CoT-decoding	**66.0** (+31.0)	**73.5** (+11.3)

Choice of k. In Figure 5, we illustrate how the choice of k, representing the number of top alternative tokens considered, influences the overall accuracy. Overall we found that higher values of k typically result in improved model performance, suggesting that in many cases, the correct CoT paths may indeed exist but are often ranked lower during model's decoding. For instruction-tuned models, the effect of k is less significant, indicating that the process of instruction-tuning effectively brings forth the majority of CoT-paths to the first few decoding paths.

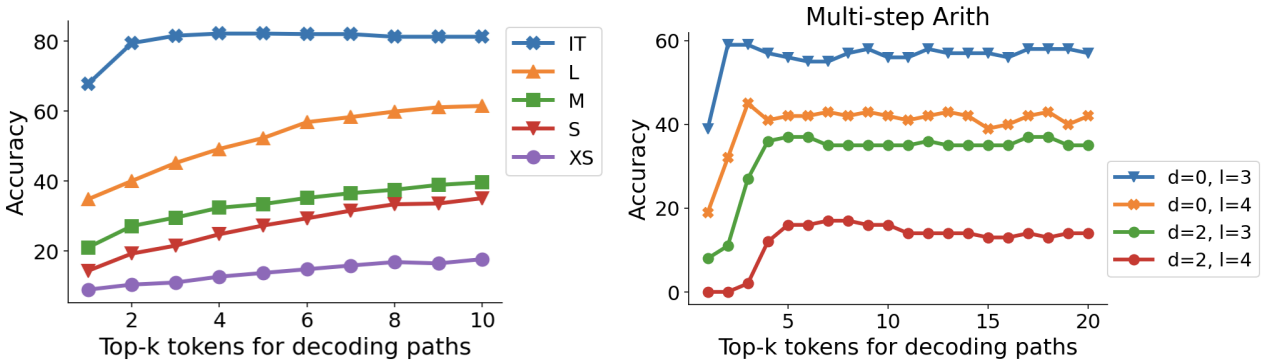

Figure 5 | The effect of k on reasoning accuracy w.r.t. PaLM-2 model scales and task difficulty.

3.2. CoT-decoding Enables a Better Understanding of Model's Intrinsic Reasoning Abilities

Compared to existing works that improve model's reasoning via better human-written prompts, a key distinction of our proposed approach lies in the complete elimination of human-provided prompts. This modification enables a *more truthful assessment* of a language model's intrinsic reasoning capabilities. In the previous section, we show that language models inherently possess reasoning capabilities for grade-school-level math problems and simple commonsense reasoning tasks. In this section, we will systematically vary the difficulty levels of synthetic tasks to gain a more comprehensive understanding of language models' inherent reasoning abilities via CoT-decoding.

We consider the following symbolic reasoning tasks: (1) the Coin Flip task from (Wei et al., 2022), with $2, 3, 4$ rounds of potential flips; and two tasks from Big-Bench-Hard (bench authors, 2023; Suzgun et al., 2022): (2) Web of lies, with $3, 4, 5$ truth/lie statements, and (3) Multi-step arithmetic with various depth level d and length l. For each task, we produce 100 examples for each difficulty level, except for Web-of-Lies (5) we use the existing dataset from (Suzgun et al., 2022). We also include two natural-language-based but synthetic tasks from Big-Bench, Sports Understanding and Object Counting, to probe model's intrinsic abilities in solving synthetic tasks.

	Coin Flip			Web of lies			Multi-step Arithmetic				Sports Und.	Object Count
	2	3	4	3	4	5	d_0, l_3	d_0, l_4	d_2, l_3	d_2, l_4		
Greedy	70.0	53.0	48.0	76.0	58.0	53.6	39.0	19.0	8.0	0.0	58.8	36.0
CoT-decoding	**94.0**	**57.0**	**55.0**	**87.0**	**63.0**	**57.6**	**56.0**	**42.0**	**35.0**	**16.0**	58.0	**39.2**

Table 6 | The model's intrinsic reasoning ability varies depending on the task difficulty levels.

The presence of correct CoT paths depends on the task difficulty levels and correlates with task prominence in the pre-training distribution. The results in Table 6 (on PaLM-2 L) show that despite CoT-decoding helps elicit better reasoning across almost all tasks, the gains vary significantly depending on the task difficulty level: the simpler the task is, the better chance that a correct reasoning path can be found. We also looked at the model's top-k decoding paths, and found that models can generate the correct CoT paths when the solution involves at most 1 or 2 step knowledge manipulation, and the model starts to struggle with generating the correct CoT-paths when the steps become 3 or more. See Figure 5 (right) where the model's accuracy improves only for larger k's as task complexity increases (higher d and l's). This phenomenon suggests that the correct CoT-paths become harder to find when the task becomes more synthetic. This mirrors the finding in (McCoy et al., 2023), where the authors show language models are highly influenced by the distribution they have been trained on.

CoT-decoding unveils model's intrinsic vulnerabilities in reasoning. Our results also unveil the specific areas where language models still struggle with: for example, on Coin-Flip and Web-of-Lies, we observe that the model can generate CoT paths that simulate the process step-by-step, but it can easily lose track of the states, especially when the task complexity increases. This reveals model's intrinsic vulnerability in performing accurate state tracking. On Multi-step Arithmetic, we observe that the model tends to perform calculations from left to right in the CoT-decoding paths, rather than following the correct mathematical order. These observations point to future directions where we should improve the models on.

In addition, over these synthetic tasks, we found that existing CoT prompts on Big-Bench-Hard (Suzgun et al., 2022) play a larger "teaching" role in guiding the model to solve such tasks, and in most cases the model just mimics the patterns in the CoT prompts to generate the correct response: e.g., the few-shot CoT prompts teach the model to perform *explicit* state tracking in each step for Web-of-lies. On the Sports Understanding task, CoT-decoding can better reveal LLMs' intrinsic strategy in solving a problem (see Appendix A), without being influenced by the external prompts which could be biased by the prompt designers. In contrast, few-shot CoT prompting constrains the model to follow an artificial strategy curated through human knowledge and intervention.

3.3. Combining CoT-decoding with CoT-Prompting

We further show that CoT-decoding can be easily combined with CoT-prompting, yielding even larger reasoning gains over multiple language models (Table 7). CoT-decoding maintains a strong performance compared to self-consistency (Wang et al., 2023a) when both are combined with CoT-prompts. Since self-consistency aggregates over multiple paths, we also show the performance based on our path aggregation algorithm, which significantly improves the model's reasoning at a similar cost. For a fair comparison, we use $k = 10$ for all methods that require multiple decoding paths.